LINEAR ALGEBRA – MATLAB PROGRAMS

III DSA | SEMESTER – 5

1. Write a matlab program on verifying basic vector space properties

Program:

```
A = [1 2 3];
B = [4 5 6];
C = [7 8 9];
k = 2;
 disp('Closure (A+B):'); disp(A+B);
disp('Commutative:');
disp(isequal(A+B, B+A));
disp('Associative:');
disp(isequal((A+B)+C, A+(B+C)));
disp('Distributive:');
disp(isequal(k*(A+B), k*A + k*B));
disp('Scalar Identity:');
disp(isequal(1*A, A));
```

2. Write a matlab program on subspace

Program:

```
u = [1 -1 0];
v = [2 -2 0];
c = 2;

if sum(u+v)==0 && sum(c*u)==0
disp('It is a SUBSPACE');
else
disp('Not a subspace');
end
```

3. Write matlab program to determine whether the given vectors form a basis of a vector space by checking their linear independence using the rank of the matrix.

Program:

```
A = input('Enter matrix whose columns are the given vectors: ');
r = rank(A);
n = size(A,2);

disp('Matrix A = ');
disp(A);

if r == n
    disp('Vectors are linearly independent');
    disp('Hence, they form a basis for their span');
else
    disp('Vectors are NOT linearly independent');
    disp('Hence, they do NOT form a basis');
end
```

4. Write a matlab program on nullspace.

Program:

```
A = [1 2 3;
     4 5 6;
```

```
     7 8 9];

disp('Matrix A = ');
disp(A);
N = null(A);
disp('Null Space of A = ');
disp(N);
```

5. Write a matlab program on Row space.

Program:

```
A = [1 2 3;
     4 5 6;
     7 8 9];

disp('Matrix A = ');
disp(A);
R = rref(A);

disp('Row Space (basis vectors) = ');
disp(R);
```

6. Write a matlab Program on Column Space

Program:

```
A = input('Enter matrix A (e.g., [1 2 3; 4 5 6; 7 8 9]): ');
[R, pivotCols] = rref(A);
col_space = A(:, pivotCols);
disp('Pivot Columns Indices:');
disp(pivotCols);
disp('Basis for Column Space:');
disp(col_space);
```

7. Write a matlab Program on dimension of a matrix.

Program:

```
A = input('Enter matrix A (e.g., [1 2; 3 4]): ');
[rows, cols] = size(A);
r = rank(A);
fprintf('Number of rows = %d\n', rows);
fprintf('Number of columns = %d\n', cols);
fprintf('Dimension of column space (Rank) = %d\n', r);
```

8. Write a matlab Program on eigen vectors and eigen values

Program:

```
A = input('Enter square matrix A (e.g., [2 1; 1 2]): ');
[rows, cols] = size(A);
if rows ~= cols
    disp('Matrix must be square');
else
    [V, D] = eig(A);
    disp('Eigenvalues (Diagonal Matrix D):');
    disp(D);
    disp('Eigenvectors (Columns of V):');
    disp(V);
end
```

9. Write a matlab Program on verifying CAYLEY HAMILTON theorem

Program:

```
A = input('Enter square matrix A (e.g., [2 1; 1 2]): ');
[n, m] = size(A);
if n ~= m
    disp('Matrix must be square');
else
    p = poly(A);
    result = zeros(n);
    for i = 1:length(p)
       power = n - (i - 1);
        if power > 0
            result = result + p(i) * (A^power);
        else
            result = result + p(i) * eye(n);
        end
    end
    disp('Result of substituting A into its characteristic equation:');
    disp(result);
    if norm(result) < 1e-6
        disp('Cayley-Hamilton theorem verified (Result ≈ Zero Matrix)');
    else
        disp('Verification failed');
    end
end
```

10. Write a matlab Program on diagonalization of a matrix

Program:

```
A = input('Enter the square matrix A: ');
[n, m] = size(A);

if n ~= m
    disp('Matrix must be square for diagonalization.');
    return;
end

[P, D] = eig(A);
disp('Matrix A:');
disp(A);
disp('Eigenvector matrix P:');
disp(P);
disp('Diagonal matrix D (Eigenvalues):');
disp(D);

if rank(P) == n
    A_check = P * D / P;
    disp('Verification (P * D * inv(P)):');
    disp(A_check);
    disp('Matrix is diagonalizable.');
else
    disp('Matrix is NOT diagonalizable.');
end
```

11. Write a MATLAB program to check whether given vectors are orthogonal or not.

Program:

```
A = input('Enter first vector (e.g., [1 2 3]): ');
B = input('Enter second vector (e.g., [4 5 6]): ');
if length(A) ~= length(B)
    disp('Vectors must be of the same dimension');
```

```
else
       dot_product = sum(A .* B);
   if dot_product == 0
        disp('The vectors are orthogonal');
    else
        disp('The vectors are NOT orthogonal');
    end
     fprintf('Dot Product = %d\n', dot_product);
end
```

12. Write a matlab Program on Orthonormal Vectors

Program:

```
A = input('Enter matrix (each row as a vector, e.g., [1 0 0; 0 1 0; 0 0 1]): ');
[n, m] = size(A);
isOrthonormal = true;
tol = 1e-10;

for i = 1:n
    if abs(norm(A(i,:)) - 1) > tol
        isOrthonormal = false;
        break;
    end
end

for i = 1:n
    for j = i+1:n
        if abs(dot(A(i,:), A(j,:))) > tol
            isOrthonormal = false;
            break;
        end
    end
end

if isOrthonormal
    disp('The given vectors are ORTHONORMAL');
else
    disp('The given vectors are NOT orthonormal');
end
```

13. Write a MATLAB program on Gram-Schmidt Process

Program:

```
A = input('Enter matrix (columns as vectors, e.g., [1 1; 1 -1]): ');
[n, m] = size(A);
Q = zeros(n, m);
tol = 1e-10;

for j = 1:m
    v = A(:,j);
    for i = 1:j-1
        proj = dot(Q(:,i), v) * Q(:,i);
        v = v - proj;
    end

    if norm(v) < tol
        error('Input columns are linearly dependent; Gram-Schmidt cannot form a full basis.');
    end

    Q(:,j) = v / norm(v);
end
```

```
disp('Orthonormal basis (columns):');
disp(Q);
```

14. Write a MATLAB program on orthogonal projection of vectors

Program:

```
A = input('Enter vector A (e.g., [2 3 4]): ');
B = input('Enter vector B (e.g., [1 0 1]): ');
if length(A) ~= length(B)
    disp('Vectors must be of the same dimension');
else
    proj = (dot(A,B) / dot(B,B)) * B;
    disp('Projection of A onto B is:');
    disp(proj);
end
```

15. Write a matlab program on rank of matrix

Program:

```
A = input('Enter the matrix: ');
disp('The entered matrix is:');
disp(A);
R = rank(A);
disp('Rank of the matrix is:');
disp(R);
fprintf('The rank of the given matrix = %d\n', R);
```

16. Write a matlab program on orthogonal set.

Program:

```
v1 = input('Enter first vector: ');
v2 = input('Enter second vector: ');
d = dot(v1,v2);
if d == 0
disp('The vectors form an orthogonal set');
else
disp('The vectors do not form an orthogonal set');
end
```

17. Write a matlab program on orthonormal set

Program:

```
v1 = input('Enter first vector: ');
v2 = input('Enter second vector: ');
if dot(v1,v2)==0 && norm(v1)==1 && norm(v2)==1
    disp('The vectors form an orthonormal set');
else
    disp('The vectors do not form an orthonormal set');
end
```

18. Write a matlab program to construct a diagonal matrix

Program:

```
n = input('Enter the order of the matrix: ');
v = input('Enter the diagonal elements as a vector: ');
```

```
if length(v) ~= n
    error('The number of diagonal elements must equal the order of the matrix.');
end

D = diag(v);

disp('Input vector is:');
disp(v);
disp('Constructed diagonal matrix is:');
disp(D);
fprintf('Order of the diagonal matrix is %d x %d\n', n, n);
disp('Verification Successful: It is a diagonal matrix.');
```

19. Write a matlab program on basis.

Program:

```
A = input('Enter the matrix A: ');
[R, pivotCols] = rref(A);
Basis = A(:,pivotCols);

disp('The given matrix is:');
disp(A);
disp('Basis of the column space is:');
disp(Basis);
```

20. Write a matlab program on change of basis

Program:

```
B = input('Enter the basis matrix B: ');
v = input('Enter the vector v: ');
C = inv(B) * v;      % Coordinates in the new basis
disp('Basis matrix B = ');
disp(B);
disp('Vector coordinates in the new basis are:');
disp(C);
fprintf('Change of basis completed.\n');
```

21. Write a matlab program on linearly independent set

Program:

```
A = input('Enter the matrix whose columns are the vectors: ');
r = rank(A);
n = size(A,2);
if r == n
    disp('The set of vectors is linearly independent');
else
    disp('The set of vectors is linearly dependent');
end
```

22. Write a matlab program on linear transformation

Program:

```
T = input('Enter the transformation matrix T: ');
v = input('Enter the input vector v: ');
Tv = T * v;   % Linear transformation
disp('Transformation matrix T = ');
disp(T);
disp('Input vector v = ');
disp(v);
```

```
disp('Transformed vector Tv = ');
disp(Tv);
```

23. Write a matlab program on inner product space

Program:

```
u = input('Enter first vector u: ');
v = input('Enter second vector v: ');
ip = dot(u,v);          % Inner product
disp('First vector u = ');
disp(u);
disp('Second vector v = ');
disp(v);
fprintf('Inner product <u,v> = %f\n', ip);
if ip == 0
    disp('Vectors are orthogonal in the inner product space');
end
```

24. Write a matlab Program on eigen vectors and eigen values

Program:

```
A = input('Enter square matrix A (e.g., [0 -1; 1 0]): ');
[n, m] = size(A);

if n ~= m
    disp('Matrix must be square');
else
    [V, D] = eig(A);
    eigenvalues = diag(D);

    disp('Eigenvalues are:');
    disp(eigenvalues);
    disp('Eigenvectors (columns of V):');
    disp(V);

    figure;
    plot(real(eigenvalues), imag(eigenvalues), 'ro', 'MarkerSize', 8, 'LineWidth', 2);
    grid on;
    xlabel('Real Axis');
    ylabel('Imaginary Axis');
    title('Eigenvalues in Complex Plane');
    hold on;
    xline(0);
    yline(0);
end
```

25. . Write a matlab program on linearly dependent set

Program:

```
A = input('Enter the matrix whose columns are the vectors: ');
r = rank(A);
n = size(A,2);
if r < n
    disp('The set of vectors is linearly dependent');
else
    disp('The set of vectors is linearly independent');
end
```